Integrated Management of the Cazones River Basin through Ecological Development

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Background

The Cazones River basin, spanning the states of Hidalgo, Puebla, and Veracruz (Mexico), is vital for water supply, agriculture, biodiversity, and local economies. However, it faces water scarcity and pollution caused by wastewater discharges and solid waste accumulation. Traditional water monitoring methods are often limited in temporal and spatial resolution, making integrated management difficult.

Motivations

This project seeks to promote sustainable and intelligent watershed management through:

- Integration of remote sensing and field data for real-time water quality assessment.
- Promotion of ecological restoration and environmental education.
- Strengthening of local partnerships for environmental monitoring and decision-making.

Framework and Index Formulas

The first stage focused on identifying priority municipalities using PCA analysis. Sampling was conducted in Villa de Ávila Camacho, Xicotepec de Juárez, Puebla, during the 2025 dry season.
Data integration included:

- In situ water quality measurements (pH, conductivity, turbidity, etc.).
- Multispectral drone imagery, generating indices such as:

Index	Bands *Formulas*
NDWI (Normalized Difference Water Index)	(GREEN - NIR) / (GREEN + NIR)
Detects surface water presence	
NDVI (Normalized Difference Vegetation Index)	(NIR - RED) / (NIR + RED)
Identifies vegetation healt	
NDTI (Normalized Difference Turbidity Index)	(RED - GREEN) / (RED + GREEN)
Estimates relative turbity	

These indices supported spatial analysis of water bodies and surrounding vegetation.

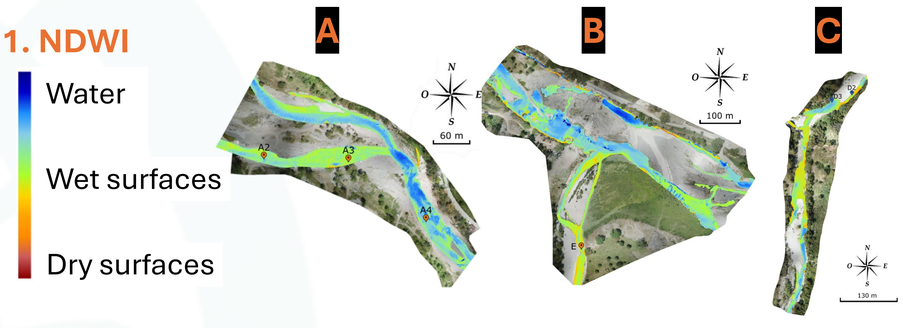
River monitoring zones



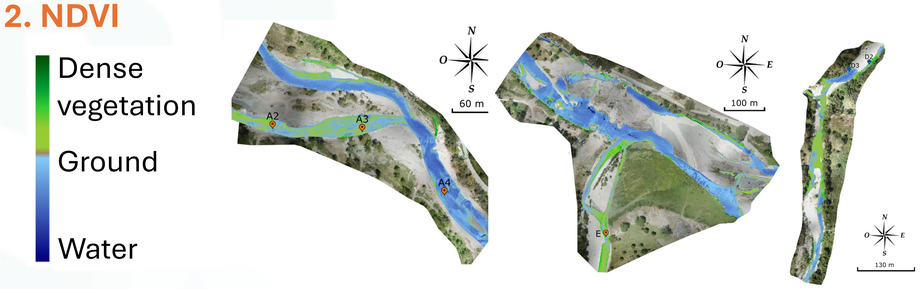
Highlights

Water quality in the San Marcos River and its tributary (Metate River) remained within acceptable limits, with no signs of severe contamination.
Multispectral imagery effectively identified water and vegetation patterns, supporting targeted field sampling.
The approach demonstrates the potential of drone-based monitoring for watershed management and ecological planning.
Collaboration with Universidad Tecnológica de Xicotepec de Juárez and local authorities strengthened the regional monitoring network.

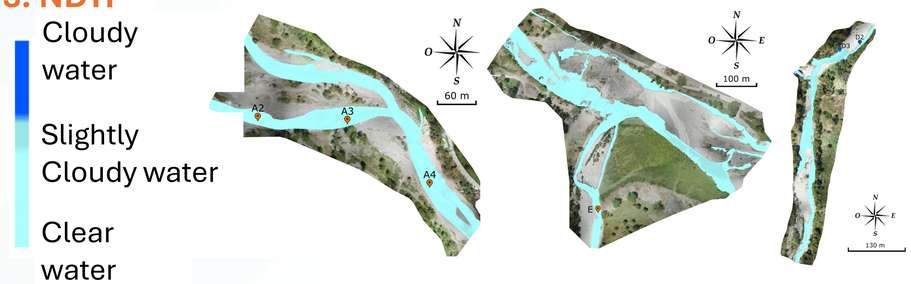
1. NDWI



2. NDVI



3. NDTI



Acknowledgements

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Reference:
Chen, B. et al. (2021). *Ecological Indicators*, 133, 108434. https://doi.org/10.1016/j.ecolind.2021.108434

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